

fMRI Analysis of Cognitive Control Using Flanker Task

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Abstract—This study presents a full fMRI analysis pipeline applied to the NYU Slow Flanker dataset (ds000102, $N = 26$) to investigate the neural substrates of cognitive control and interference resolution. Using the Eriksen Flanker paradigm where participants respond to a central target arrow amid congruent or incongruent flanking distractors, we mapped BOLD signal differences between conditions that vary in cognitive demand. Preprocessing was conducted in FSL and included skull-stripping (BET), motion correction (MCFLIRT), Gaussian spatial smoothing, and two-step registration to MNI152 standard space. A three-level GLM framework was employed: first-level modeling of individual runs, second-level fixed-effects aggregation within subjects, and third-level population inference using FLAME 1. The critical Incongruent > Congruent contrast identified eight significant clusters ($Z > 3.1$, $p < 0.05$), with peak activation in the right superior lateral occipital cortex (379 voxels, $Z = 4.28$), alongside engagement of the inferior frontal gyrus, paracalcarulate gyrus, and insular cortex. ROI analysis using both anatomical and spherical masks confirmed that incongruent trials consistently elicited greater BOLD responses across all targeted regions. These findings delineate a distributed fronto-occipital network underlying visual conflict resolution, response inhibition, and conflict monitoring, providing a reproducible neuroimaging template for studying executive function.

Index Terms—fMRI, Flanker Task, FSL, FEAT, BOLD, General Linear Model, Cognitive Control, MCFLIRT, BET, MNI152, Neuroimaging, BIDS.

I. INTRODUCTION

Understanding the neural architecture underlying human attention and executive function remains a central objective in cognitive neuroscience. Functional MRI (fMRI) has emerged as the predominant tool for non-invasively mapping these processes, exploiting the tight neurovascular coupling between neural activity and haemodynamic blood flow. [1]

The Eriksen Flanker task provides a well-validated experimental model for probing response conflict and cognitive interference resolution. Participants must respond to a central target arrow while ignoring flanking distractors that are either congruent (pointing the same direction) or incongruent (pointing the opposite direction). Incongruent trials reliably produce slower reaction times and greater activation in fronto-parietal control networks. [3]

The present work describes a complete, reproducible analysis pipeline applied to the NYU Slow Flanker dataset (ds000102) using FSL. We detail the data organisation strategy, quality-control metrics, preprocessing stages, GLM formulation, and statistical thresholding approach, providing a methodological template for fMRI cognitive studies.

II. MATERIALS & METHODS

A. Experimental Design & Dataset Description

The analysis utilizes the NYU Slow Flanker dataset, organized according to the Brain Imaging Data Structure (BIDS) standard. [2]

TABLE I: MRI Acquisition Parameters

Parameter	Anatomical (T1w)	Functional (EPI)
Sequence	3D MPRAGE	2D EPI
TR	2500 ms	2000 ms
TE	3.93 ms	30 ms
Flip Angle	8°	80°
Voxel size	1×1×1 mm	3×3×4 mm
Volumes/Run	—	146

Each participant contributed one high-resolution T1-weighted anatomical scan and two functional task runs (146 volumes each). Behavioral task logs recorded 24 trials per run (12 congruent, 12 incongruent) with trial onset, duration, and type.

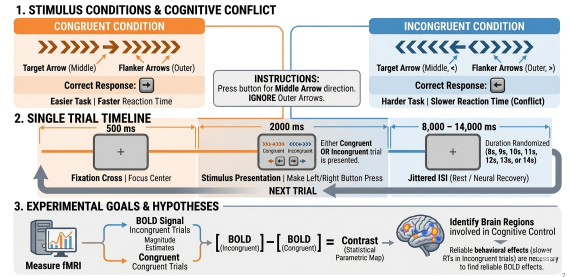


Fig. 1: Illustration of the Flanker task. Each trial begins with a fixation cross, followed by a Congruent or Incongruent stimulus (2000 ms), adapted from Kelly et al. (2008). [2]

B. Data Organization and BIDS

All data were organised according to the Brain Imaging Data Structure (BIDS) standard, ensuring software compatibility with FSL and automated validation. The directory hierarchy separates each subject's anatomical (anat/) and functional (func/) data, with paired JSON sidecar files storing scanner metadata alongside NIfTI images. BIDS compliance enables seamless ingestion by FSL preprocessing pipelines and downstream group-level toolboxes without manual path specification. [6]

C. Quality Control

Comprehensive quality control was performed to quantify image clarity, artifact presence, and subject motion.

1) Anatomical Assessment (aMRIQC)

The T1-weighted images exhibited robust signal integrity, with Signal-to-Noise Ratios (SNR) ranging from 13.7 to

over 18.7. The Maximum Entropy Focus Criterion (EFC) was 0.56, indicating minimal head motion and ghosting.

2) Functional Assessment (fMRIQC)

Time-series data demonstrated outstanding temporal stability. The maximum mean Framewise Displacement (FD) across the cohort was 0.133 mm, well below the conventional 0.20 mm threshold for exclusion.

D. Preprocessing

To prepare the raw BOLD data for statistical modeling, a rigorous fMRI preprocessing pipeline was implemented using FSL:

1) Brain Extraction (Skull-stripping)

The Brain Extraction Tool (BET) was applied to anatomical images. Based on visual inspection, an optimal fractional intensity threshold (f) between 0.2 and 0.3 was selected to discard the skull while preserving the cortex.

2) Motion Correction

FSL's MCFLIRT was used to realign all volumes to a definitive reference volume (the middle volume) to correct for spatial blurring and artificial signal variance caused by subject head motion.

3) Slice-Timing Correction

Temporal discrepancies arising from sequential slice acquisition (40 slices acquired over the 2 s TR) are accounted for in the first-level GLM via inclusion of temporal derivatives. This approach avoids unnecessary interpolation of raw BOLD time series data.

4) Smoothing

A Gaussian filter (measured in Full Width at Half Maximum or FWHM) was applied to average the fMRI signal across neighboring voxels, reducing high-frequency noise and boosting SNR.

5) Registration & Normalization

A two-step alignment pipeline mapped functional data to standard space. First, functional EPI images were aligned to the subject's high-resolution T1 image using a 12 Degree-of-Freedom (DOF) affine transformation. Second, the anatomical image was normalized to the standard MNI152 template, allowing for cross-subject comparison.

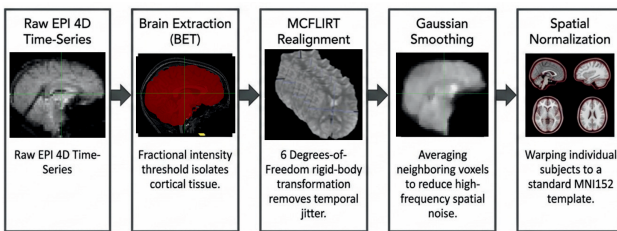


Fig. 2: Standard fMRI preprocessing pipeline.

E. First-Level Statistical Analysis

The first-level analysis employs the General Linear Model (GLM) to statistically evaluate task-evoked brain activity for each participant, enabling the identification of regions significantly activated during cognitive control processes.

1) General Linear Model Formulation

For each voxel, the measured BOLD time-series Y (size $T \times 1$, $T = 146$ volumes) was modeled as:

$$Y = X\beta + \epsilon \quad (1)$$

where X ($T \times p$) is the design matrix containing predicted HRF-convolved regressors, β ($p \times 1$) is the vector of parameter estimates (beta weights), and ϵ ($T \times 1$) is the residual noise. Parameters were estimated using ordinary least squares:

$$\hat{\beta} = (X^T X)^{-1} X^T Y \quad (2)$$

2) Contrasts and Statistical Inference

Three contrasts were defined:

- **Contrast 1:** Incongruent > baseline ($c = [1, 0]$)
- **Contrast 2:** Congruent > baseline ($c = [0, 1]$)
- **Contrast 3:** Incongruent > Congruent ($c = [1, -1]$)

For each contrast, the contrast of parameter estimate (COPE) was computed as

$$\hat{c} = c^T \hat{\beta}, \quad (3)$$

with variance

$$\text{Var}(\hat{c}) = c^T (X^T X)^{-1} c \cdot \hat{\sigma}^2, \quad (4)$$

where $\hat{\sigma}^2$ is the estimated residual variance. The t-statistic was then:

$$t = \frac{\hat{c}}{\sqrt{\text{Var}(\hat{c})}} \quad (5)$$

3) Model Fitting

The GLM mathematically compares the expected time-series against the preprocessed raw fMRI data. This approach partitions the observed signal variance into components explained by the experimental design (the predicted BOLD response) and residual error (baseline noise, physiological artifacts, and unexplained variance). For each voxel, the model estimates beta weights (β) representing the magnitude of BOLD signal contribution for each trial type. Temporal autocorrelation was addressed using pre-whitening to ensure valid statistical inference.

F. Second-Level fMRI Analysis: Within-Subject Aggregation

Following first-level single-run modeling, a second-level analysis is required for study designs involving multiple scanning sessions (runs) per participant. The primary objective of this intermediate computational step is to aggregate the lower-level Contrast of Parameter Estimates (COPEs) — such as task-specific condition subtractions — within each individual subject. This aggregation increases the intra-subject signal-to-noise ratio and condenses the data prior to group-level population inference.

1) Statistical Modeling Framework

Intra-subject aggregation is executed utilizing a Fixed Effects statistical model. While higher-level group analyses typically require Mixed Effects models (e.g., FSL's FLAME 1 or ordinary least squares) to account for inter-subject variance and generalize findings to a wider population, the Fixed Effects model is the mathematical standard for within-subject averaging. It calculates the mean of the parameter estimates across a single subject's runs without attempting to model cross-session random variance.

2) General Linear Model (GLM) Specification

GLM matrix configuration: the 52×26 sparse EV matrix with diagonal contrasts (2nd level, within-subject fixed effects) produces 26 averaged statistical maps — one per subject.

G. Third-Level fMRI Analysis: Population Inference and Generalization

While second-level analysis aggregates data within individual subjects, a third-level analysis is required to generalize the functional findings from the specific study sample ($N = 26$ participants) to the broader population. Within the FSL framework, this group-level evaluation computes the mean and standard error of specific contrast estimates across all participants to determine the statistical significance of population-level cortical activation.

1) Data Inputs and GLM Specification

The direct inputs for the third-level FEAT pipeline were the 3D Contrast of Parameter Estimates (COPE) images generated from the second-level within-subject aggregation. Specifically, cope3 images, representing the subtraction of functional networks (e.g., Incongruent minus Congruent tasks), were utilized. The GLM was constructed using a “single group average” experimental design. The design matrix consisted of 26 input rows (one per subject) and a single Explanatory Variable (EV1) populated entirely by ones (1.0).

2) Variance Modeling and Statistical Inference

In Stat Inference, we have several options: Fixed Effects, Mixed Effects: Simple OLS, Mixed Effects: FLAME 1, Mixed Effects: FLAME 1+2, and Randomise. But we will proceed with **Mixed Effects: FLAME 1**: The standard robust model for population inference, which weights each subject’s data inversely to their intra-subject variance (i.e., noisier subject data is mathematically de-weighted).

3) Post-Statistical Thresholding

Following the generation of group-level Z-statistic spatial maps, rigorous topological thresholding was applied to differentiate true biological Blood-Oxygen-Level-Dependent (BOLD) signals from mathematical noise. Thresholding comparison: brain schematic panels for all four methods (None $\rightarrow$ Uncorrected $\rightarrow$ Voxel GRF $\rightarrow$ Cluster), with the cluster-based approach highlighted as used in this study.

H. Region of Interest (ROI) Analysis

1) Anatomical Mask Method

Z-statistics from all 26 subjects were extracted per region using FSL’s `fslmeants` and submitted to one-sample t-tests. Both LOC divisions reached significance ($p < 0.05$) in the **Incongruent–Congruent** contrast, while the PCG and Insula did not, likely due to the large spatial extent of their anatomical masks diluting the signal.

2) Spherical Mask Method

To improve spatial specificity, 5 mm-radius spherical ROIs were centered on peak activation coordinates. This approach revealed significant activation ($p < 0.001$) in the PCG and Insula — previously non-significant — confirming that earlier null results reflected mask size rather than a true absence of effect. Across all regions and both methods, the incongruent condition consistently yielded higher mean z-statistics, reflecting the greater cognitive demand of resolving conflicting flanker stimuli.

III. RESULTS

A. First Level Analysis Result

Single-subject GLM fitting produced clear parameter estimate (PE) and COPE maps. After Z-score normalization and cluster correction ($Z > 3.1$, $p < 0.05$), task-evoked

activation was evident in bilateral occipital and frontal regions for both conditions, confirming successful model fitting at the individual subject level.

B. Second Level Analysis Result

Fixed-effects averaging across runs increased intra-subject SNR. Glass-brain views of the three averaged COPE maps (Fig. 3) confirm consistent bilateral occipital and prefrontal engagement across subjects, with COPE3 (Incongruent–Congruent) revealing a focused conflict-specific network distinct from the broad activation seen in both individual conditions.

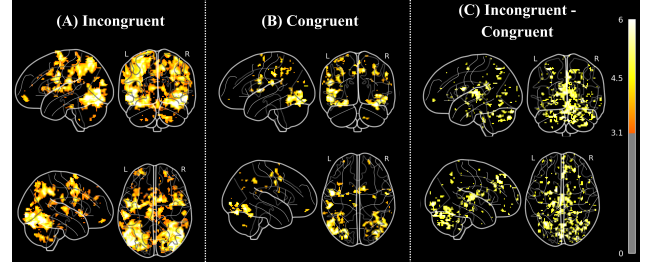


Fig. 3: Averaged contrast maps for all subjects on Glass Brain for COPE1, COPE2, and COPE3.

C. Third Level Analysis Result

Group-level FLAME1 inference on COPE3 identified eight significant clusters (Table II), spanning the lateral occipital cortex (LOC), inferior frontal gyrus (IFG), paracingulate gyrus (PCG), and insular cortex. The largest cluster (379 voxels; $p = 1.13 \times 10^{-6}$; max $Z = 4.28$) was located in the right superior LOC at MNI coordinates (26, -68 , 50). The group activation maps for all three contrasts are shown in Fig. 4.

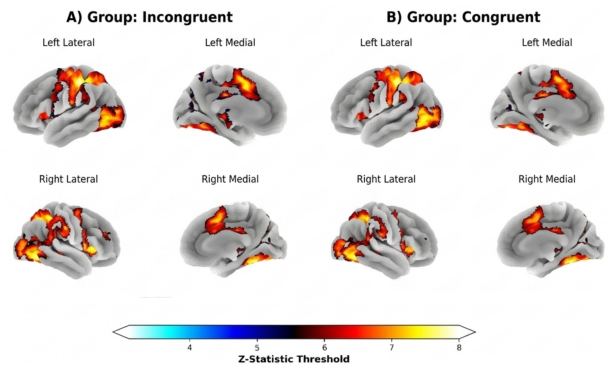


Fig. 4: Third-level group map showing activation differences for Incongruent, Congruent, and Incongruent – Congruent contrasts.

D. ROI Analysis

These results implicate a distributed network — spanning visual processing (LOC), conflict monitoring (PCG), and salience detection (Insula) — differentially engaged by the incongruent condition of the Flanker task. Across all regions and both methods, the incongruent condition consistently yielded higher mean Z-statistics (Fig. 5).

TABLE II: Cluster-level results and atlas-based anatomical localization for the Incongruent > Congruent contrast ($Z > 3.1$, $p < 0.05$).

Cluster	Voxels	p	Max Z	X	Y	Z	Atlas-based Localization	Role in Flanker Task
8	379	1.13×10^{-6}	4.28	26	-68	50	Lateral occipital cortex, superior division	Visual Spatial Attention
7	248	7.83×10^{-5}	3.96	48	-68	-14	Lateral occipital cortex, inferior division	Visual Object Recognition
6	226	1.71×10^{-4}	4.47	-44	-72	-4	Lateral occipital cortex, inferior division	Visual Object Recognition
5	211	2.96×10^{-4}	4.13	48	12	30	Inferior frontal gyrus, pars opercularis / precentral gyrus	Response Inhibition
4	127	8.43×10^{-3}	3.77	-4	16	46	Paracingulate gyrus / superior frontal gyrus	Conflict Detection
3	125	9.20×10^{-3}	3.83	-24	-64	52	Lateral occipital cortex, superior division / superior parietal lobule	Visual Spatial Attention
2	101	2.72×10^{-2}	4.49	-42	-88	-6	Lateral occipital cortex, inferior division / occipital pole	Visual Object Recognition
1	97	3.27×10^{-2}	4.06	32	22	-6	Insular cortex / frontal orbital cortex	Task Engagement / Salience

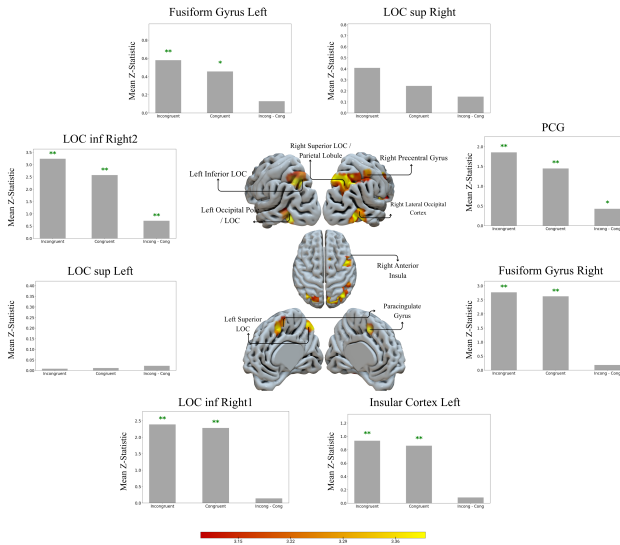


Fig. 5: Region of Interest (ROI) analysis across the Lateral Occipital Cortex (LOC, inferior and superior divisions), Paracingulate Gyrus (PCG), and Insular Cortex for all three contrasts: Incongruent (COPE1), Congruent (COPE2), and Incongruent–Congruent (COPE3).

IV. DISCUSSION

The group-level results reveal a distributed fronto-occipital network differentially recruited during incongruent compared with congruent Flanker trials, consistent with established models of cognitive control [3].

Visual processing (LOC). The dominant clusters were located bilaterally in the lateral occipital cortex (clusters 2, 3, 6, 7, 8), reflecting heightened visual discrimination demands when target and distractor arrows are spatially competing. Greater LOC engagement under incongruency aligns with early sensory filtering accounts, suggesting that conflict resolution begins at a perceptual rather than purely executive stage.

Response inhibition (IFG/precentral gyrus). Cluster 5, centred in the right IFG pars opercularis and precentral gyrus ($Z_{\max} = 4.13$), implicates the canonical response-inhibition network. The IFG is known to gate motor output and suppress prepotent responses [3], and its right-lateralized recruitment here is consistent with stop-signal

and go/no-go literature.

Conflict monitoring (PCG). Cluster 4 in the paracingulate/superior frontal gyrus represents the conflict-monitoring role classically assigned to the anterior cingulate cortex (ACC). The PCG signal was detectable only with spatially precise spherical ROIs, highlighting that coarse anatomical masks can obscure focal activations and underscoring the value of combining whole-brain and targeted ROI approaches.

Salience detection (Insula). The right insular/frontal-orbital cluster (cluster 1) reflects broader task-engagement and salience processing, consistent with the Insula’s role in interoceptive awareness and error signalling during demanding cognitive tasks.

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APPENDIX A PROJECT REPOSITORY

All scripts and processing results associated with this manuscript are publicly available in a GitHub repository: <https://github.com/RaghadAbdelhameed/fMRI-Study-of-Cognitive-Control-in-the-Flanker-Task>.
git